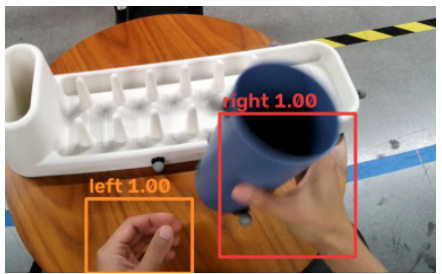
Upstream V2D hand stage (our baseline E0), and where each model sits

ego video (cam_a) → WiLoR finds each hand (box, left/right) → SAM2 hand mask in every frame → HaMeR MANO hand per frame → MoGe depth sets the hand's distance → 3D hand tracks

HaWoR (alternative, E2): one network takes the whole video clip and outputs MANO hands for every frame, plus the camera's motion; it also fills in frames where a hand is briefly hidden

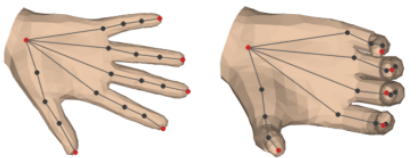
(a) WiLoR: hand boxes in cam_a



(b) HaMeR: one MANO hand per frame



(c) MANO: flat, and posed as in (b)



(d) HaWoR on HOT3D (green: true hand)

